PLL Presentation

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[redacted]

K_{VCO} = 222.22 kHz/V

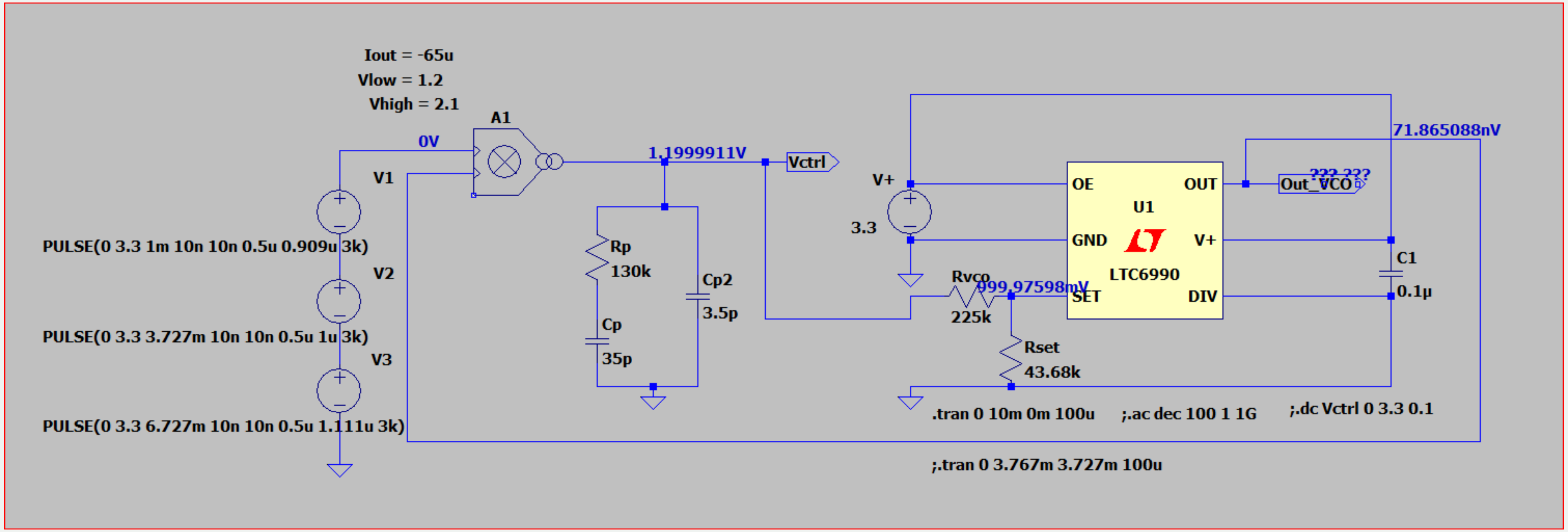
R_{VCO} = 225 $k\Omega$

R_{SET} = 43.61$k\Omega$

$Vctrl(max) = 2.1$ V

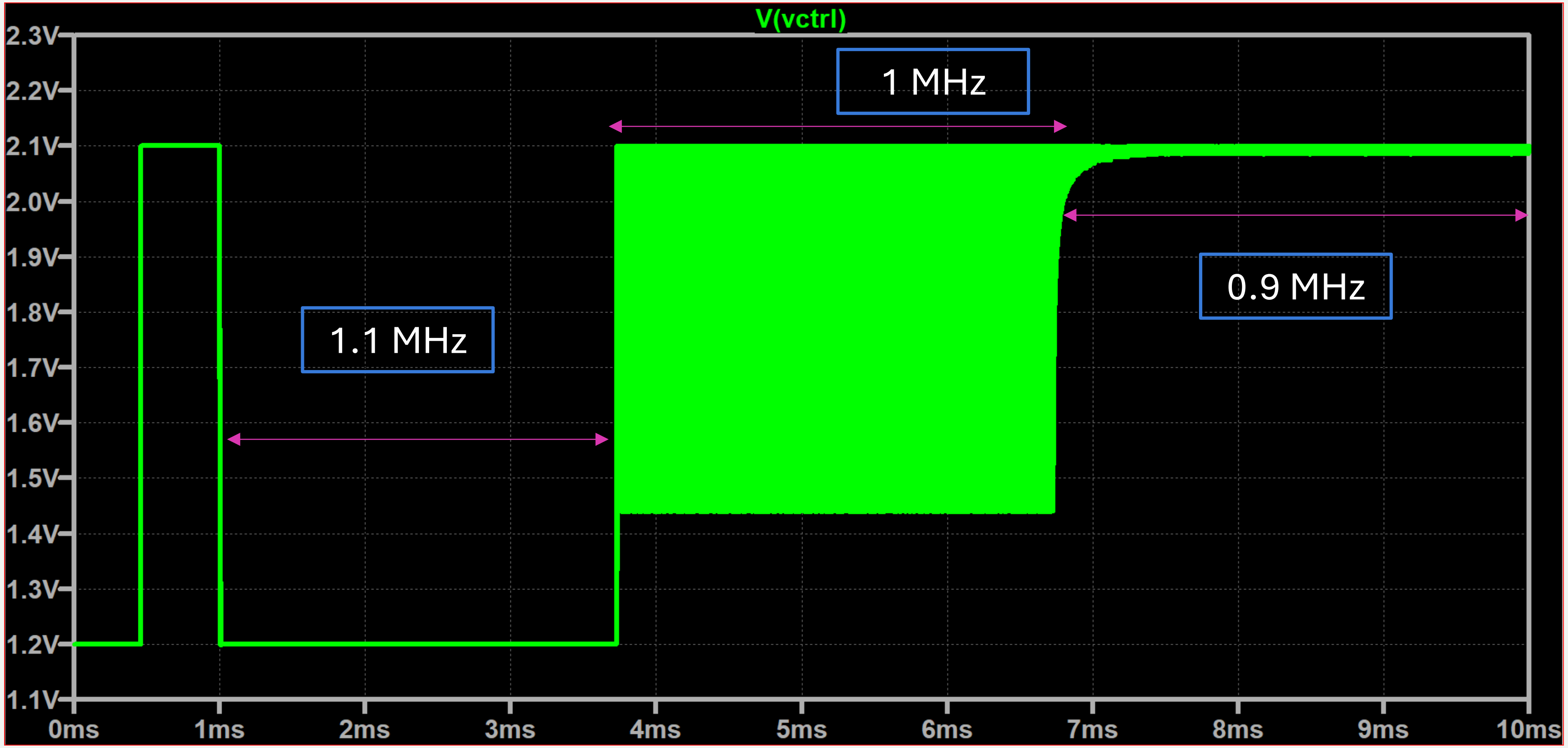
$Vctrl(min) = 1.2$ V

$K_{PD} = 10.35 \frac{\mu A}{rad/s}$



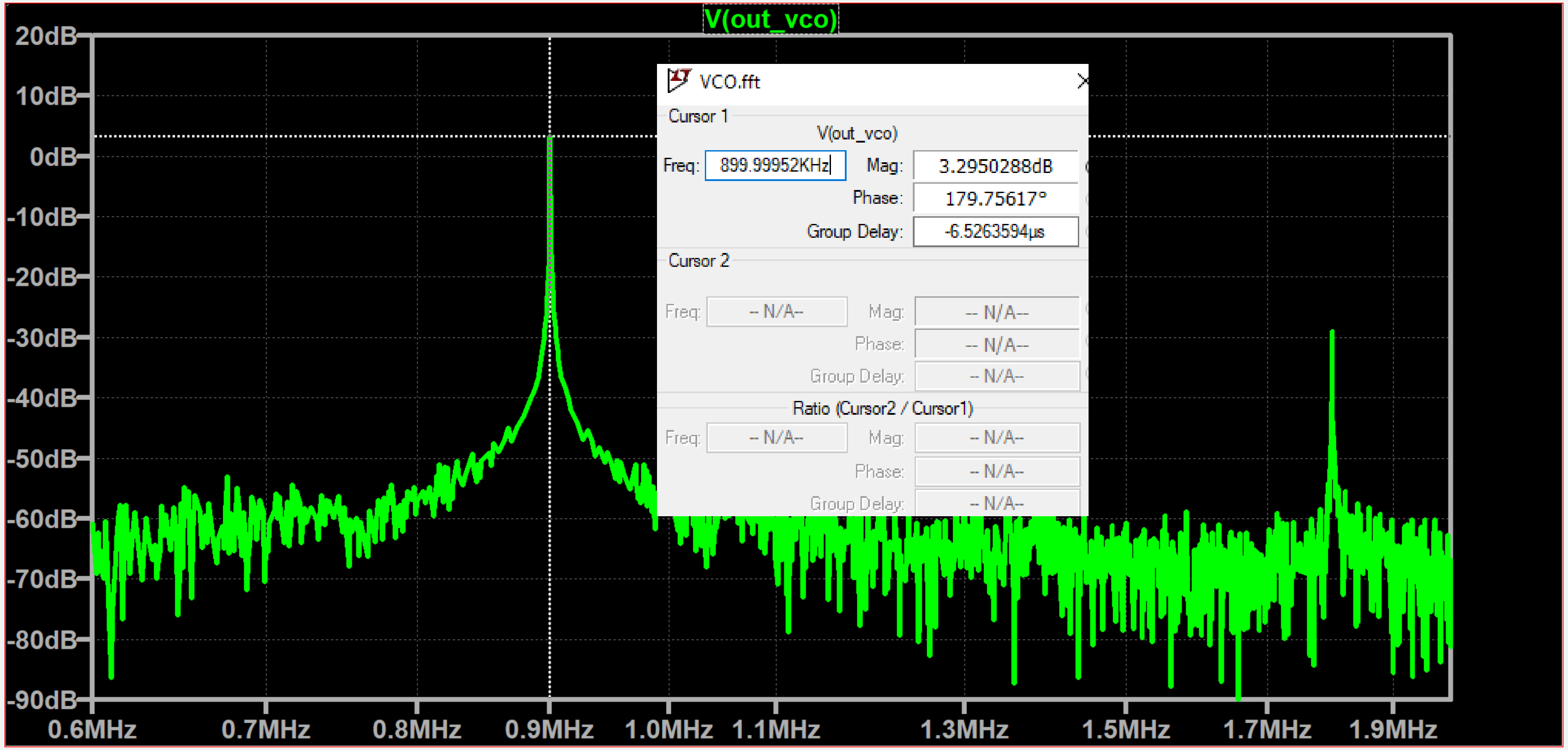
PLL Schematic

[redacted]

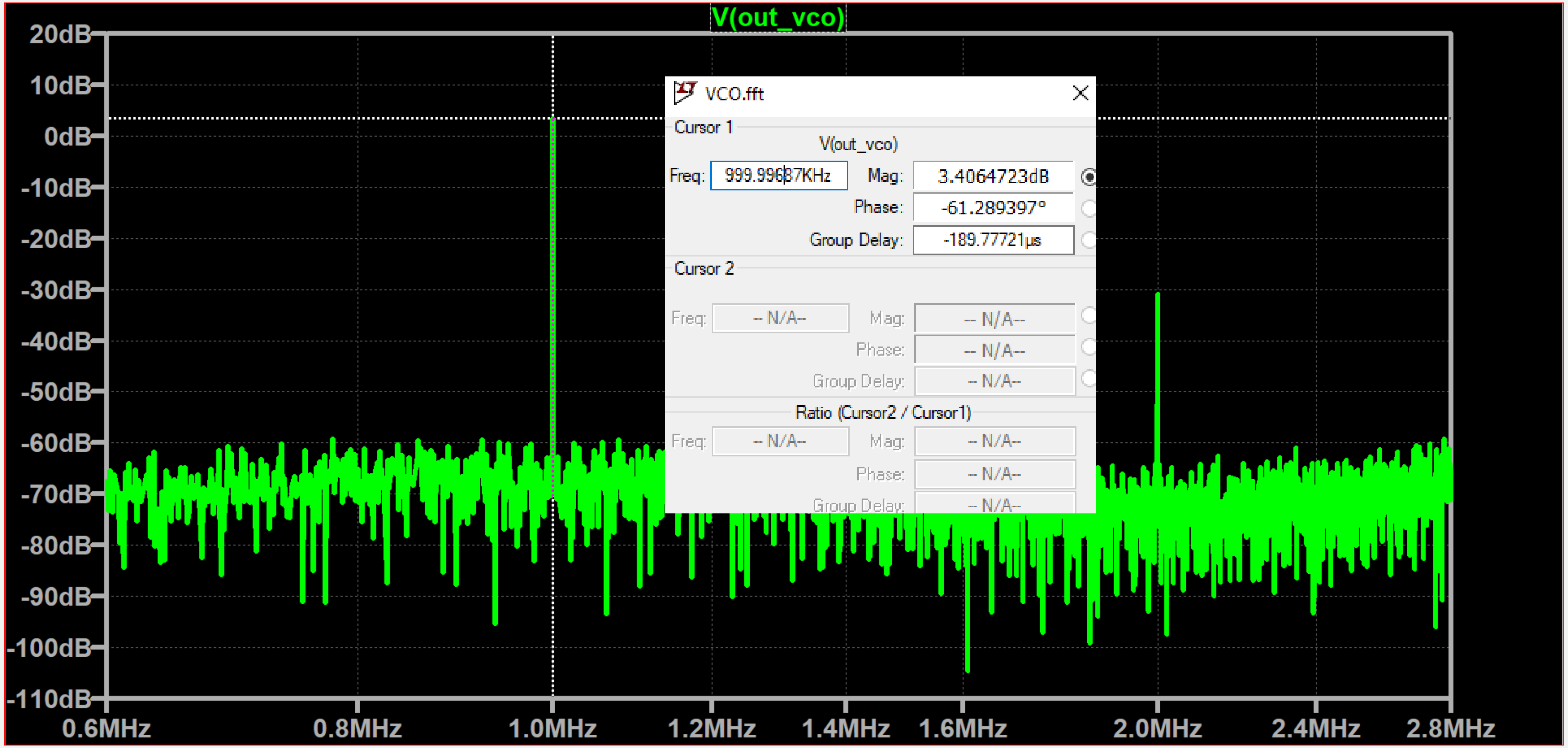


Control Voltage after using step frequency input

[redacted]

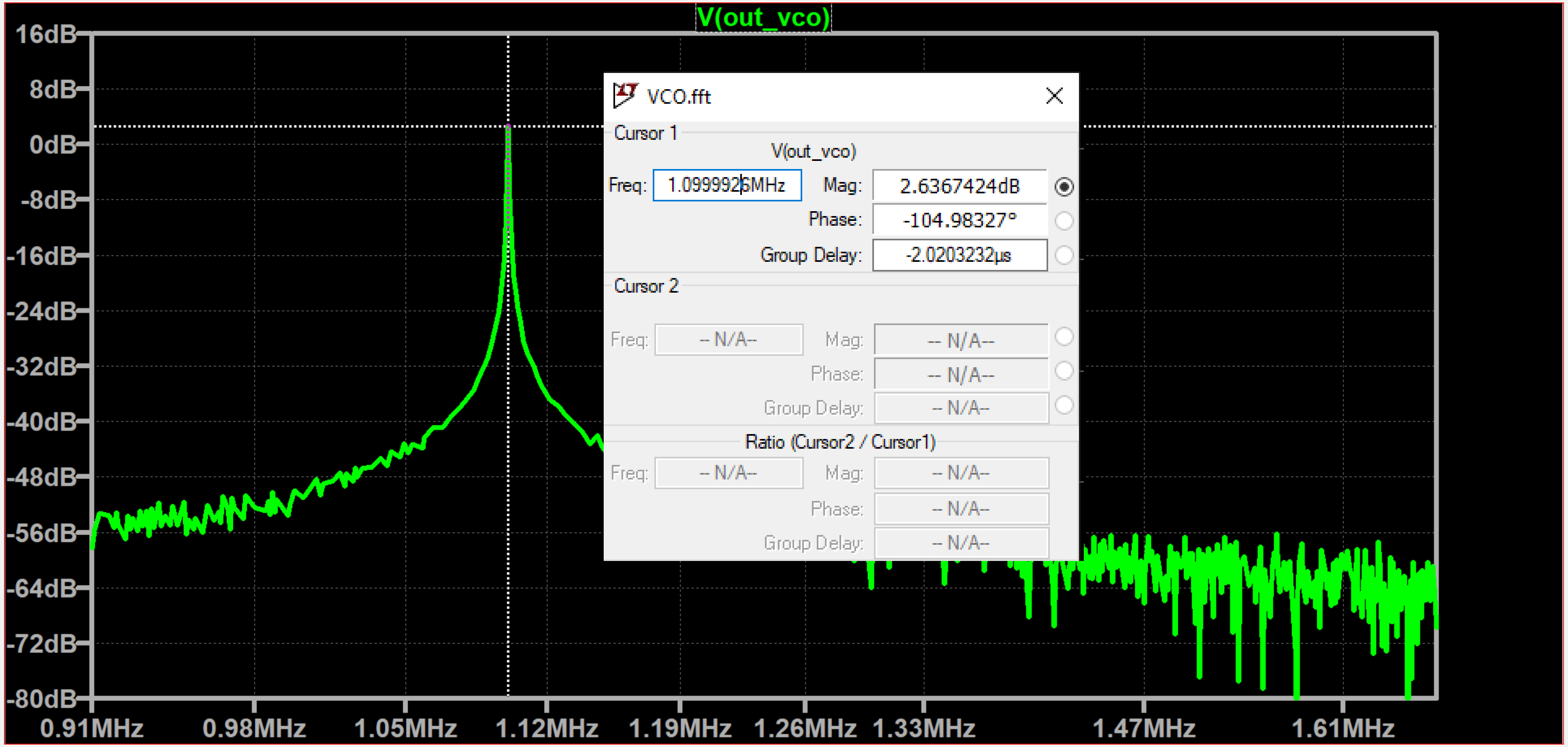


FFT of 0.9 MHz Output Voltage



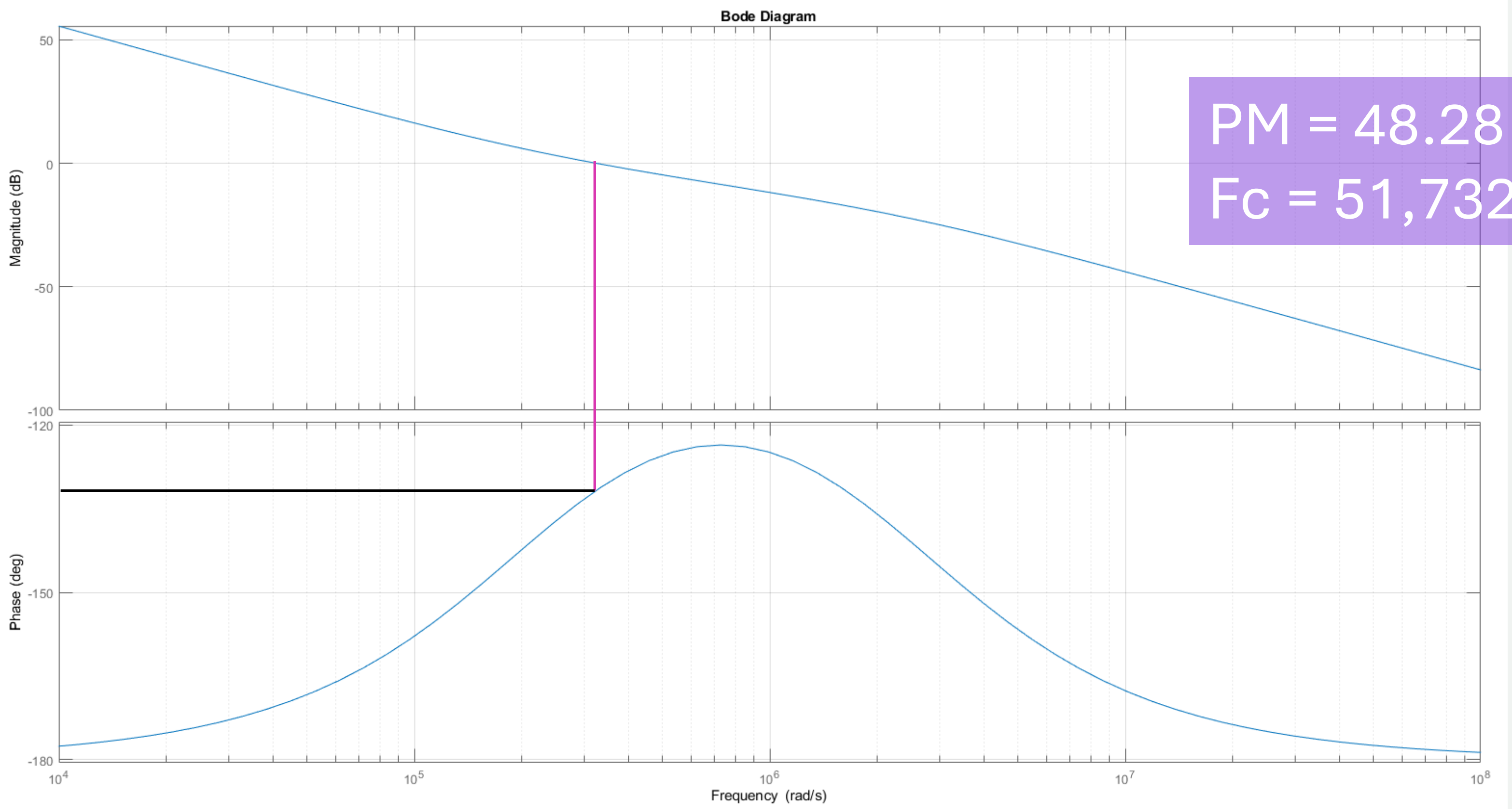
FFT of 1 MHz Output Voltage

[redacted]



FFT of 1.1 MHz Output Voltage

MATLAB Modeling



PM = 48.28 degree
Fc = 51,732 kHz

Bode plot of Theoretical modeling

MATLAB Modeling

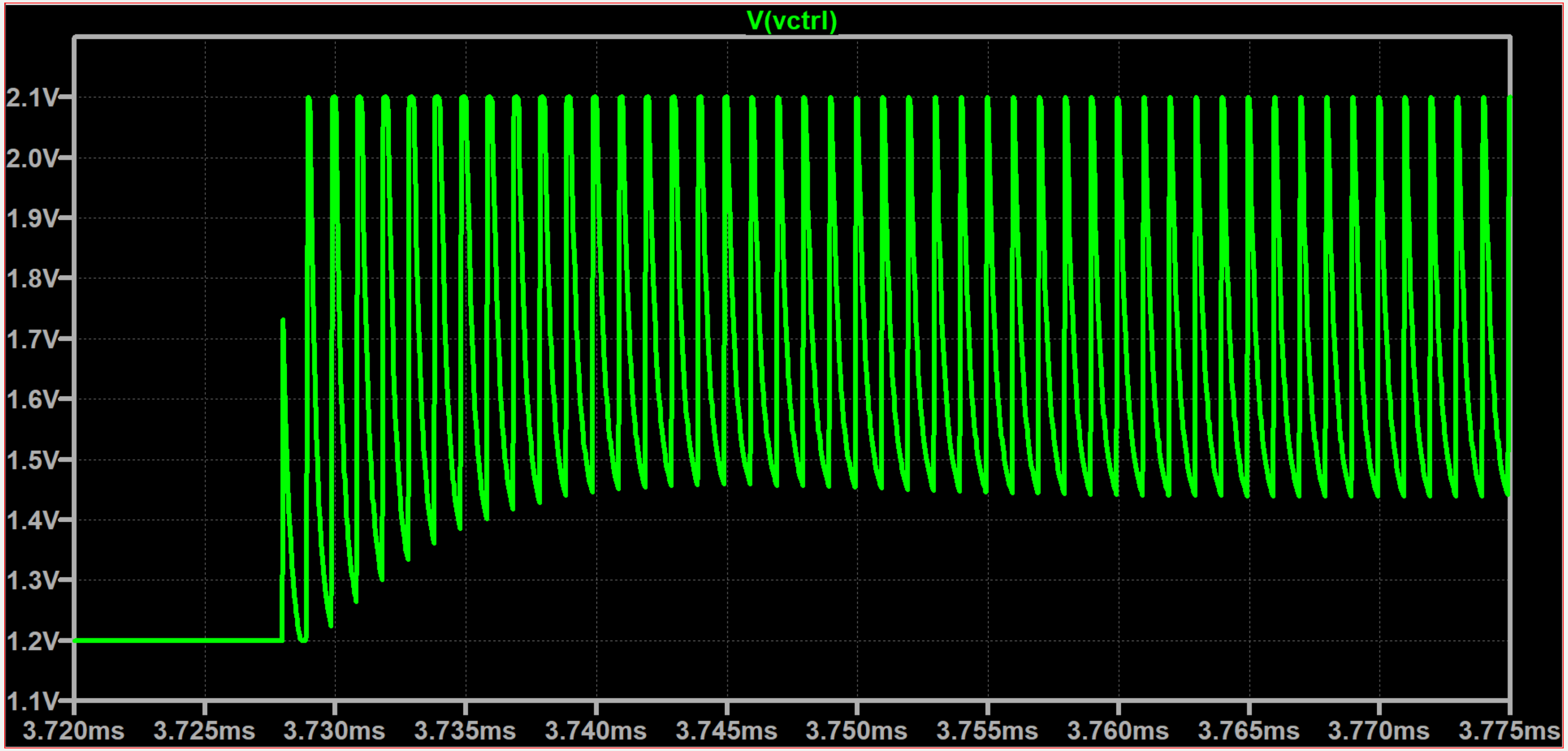
Find Phase Margin and Crossover frequency by calculating the damping ratio from Overshoot and that of natural frequency from damping ratio and settling time for finding crossover frequency

$$\xi = \frac{-\ln(\frac{OS}{100})}{\sqrt{(\pi^2 + ln^2(OS/100))}}, \qquad Ts = \frac{4}{\xi\omega n},$$

$$PM = tan^{-1}(2\xi\sqrt{(2\xi^2} + \sqrt{(4\xi^4 + 1)}))$$

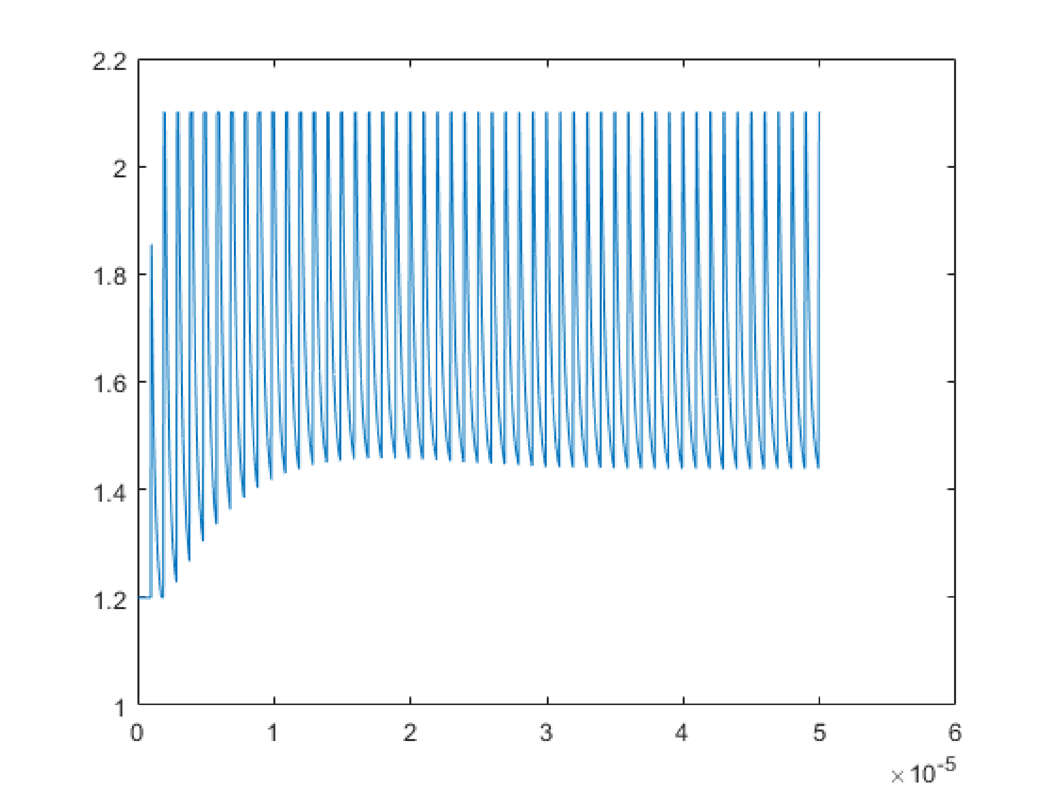
$$\omega c = \omega n * (\sqrt{(2\xi^2} + \sqrt{(4\xi^4 + 1)}))$$

[redacted]

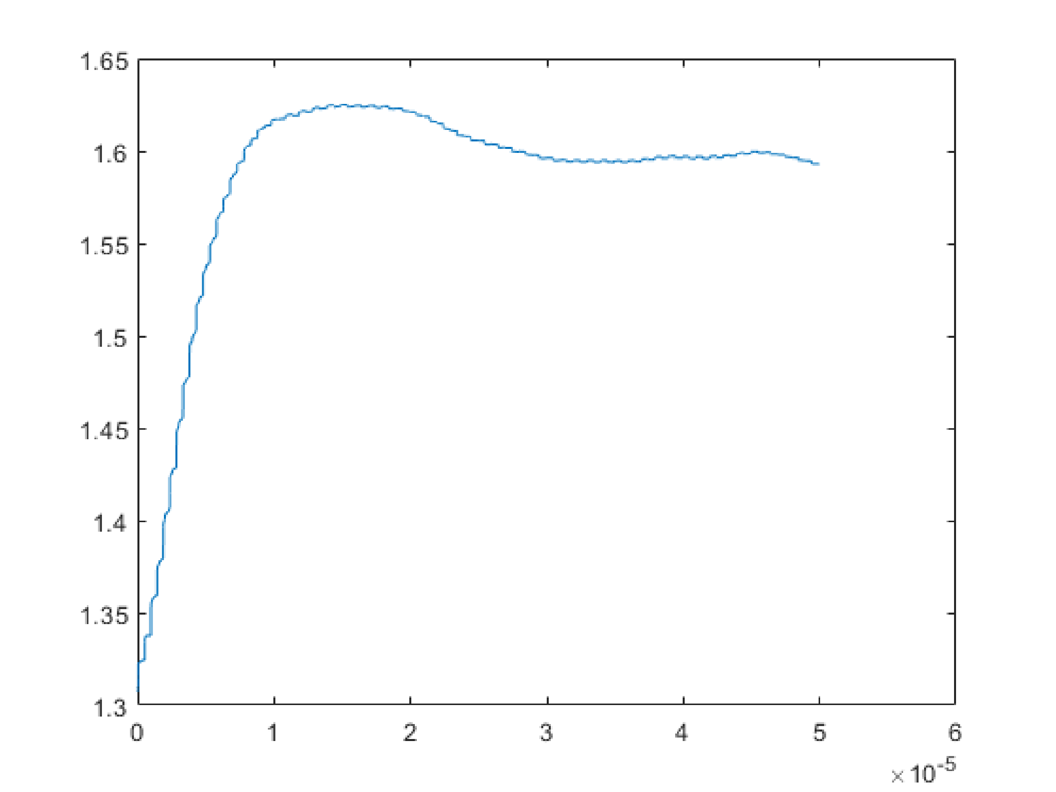


Control Voltage of 1MHz pulse wave Input

[redacted]

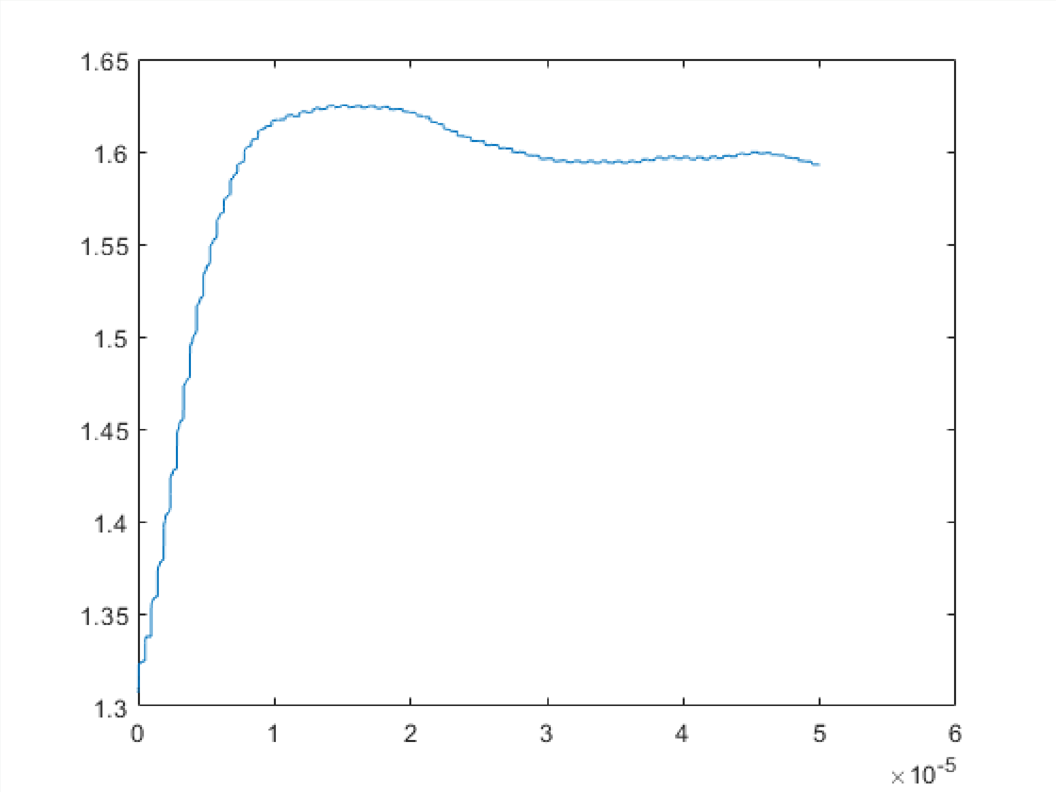


50 us time of 1 MHz step frequency



Smooth with gaussian 1400 taps

[redacted]



Smooth with gaussian 1400 taps

```
ans =

  struct with fields:

        RiseTime: 2.8397e-06
   TransientTime: 4.6421e-05
    SettlingTime: 1.5369e-05
     SettlingMin: 1.4340
     SettlingMax: 1.6250
       Overshoot: 2.0042
      Undershoot: 0
            Peak: 1.6250
        PeakTime: 1.5132e-05


Fcross =

   5.3982e-04


PM =

   69.8866
```

Summary

With Design, 2 % Overshoot and 15.37 us Settling time is achieved, resulting in 69.89 Phase margin and 53.98 kHz Crossover frequency. PLL can be locked within 0.9 – 1.1 MHz with 53.98 kHz crossover frequency and 69.89-degree phase margin

Thank you for listening